

JOSE M. ALVAREZ

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CURRENT POSITION

Santander AI Lab

Senior Research Scientist

Mar 2026 -

Madrid, ES

EDUCATION

Scuola Normale Superiore

Ph.D. in Data Science

2020 - 2024

Advisor: [Salvatore Ruggieri](#)

Dissertation: “Causality for Fair Machine Learning”

Committee: [Mykola Pechenizkiy](#), [Joshua Loftus](#), [Eirini Ntoutsis](#)

Toulouse School of Economics

M.Sc. in Econometrics and Empirical Economics

2015 - 2017

University of Florida

B.A. in Economics, *summa cum laude*

B.A. in History

2011 - 2015

Lester B. Pearson United World College of the Pacific

International Baccalaureate

2009 - 2011

AWARDS & HONORS

Best Reviewer, International Conference on Machine Learning

2024

Marie Skłodowska-Curie Fellowship ([NoBIAS ITN](#)), European Commission

2020 - 2023

Phi Beta Kappa, University of Florida

2015

Honors Program, University of Florida

2011 - 2015

Davis United Word College Scholar (full-ride), University of Florida

2011 - 2015

United Word College Scholarship (full-ride), Lester B. Pearson UWC

2009 - 2011

PUBLICATIONS & MANUSCRIPTS

Research interests: responsible AI, causal algorithmic fairness, human-in-the-loop

See [Google Scholar](#) for an up to date list; * shared first author

Journal Articles

1. **J. M. Álvarez** and S. Ruggieri. Mutatis mutandis: Revisiting the comparator in discrimination testing. *Computational Intelligence*, 42(4):e70238, 2026. [[paper](#)] [[code](#)]
2. **J. M. Álvarez** and S. Ruggieri. Counterfactual situation testing: From single to multidimensional discrimination. *J. Artif. Intell. Res.*, 82:2279–2323, 2025. [[paper](#)] [[code](#)]
3. **J. M. Álvarez**, A. Bringas-Colmenarejo, A. Elobaid, S. Fabbriizzi, M. Fahimi, A. Ferrara, S. Ghodsi, C. Mougan, I. Papageorgiou, P. Reyero, et al. Policy advice and best practices on bias and fairness in AI. *Ethics and Information Technology*, 26(2):31, 2024. [[paper](#)]

4. M. Lazzari, **J. M. Álvarez**, and S. Ruggieri. Predicting and explaining employee turnover intention. *Int. J. Data Sci. Anal.*, 14(3):279–292, 2022. [\[paper\]](#)

Conference Articles

1. L. Bothmann, P. A. Boustani, **J. M. Álvarez**, G. Casalicchio, B. Bischl, and S. Dandl. Privilege scores. In *FAccT*, pages 4293–4341. ACM, 2026. [\[paper\]](#) [\[code\]](#)
2. **J. M. Álvarez** and S. Ruggieri. Toward a causal framework for modeling perception. *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, 8(1):166–178, 2025. [\[paper\]](#)
3. F. Palomba*, A. Pugnana*, **J. M. Álvarez**, and S. Ruggieri. A causal framework for evaluating deferring systems. In *AISTATS*, pages 2143–2151. PMLR, 2025. [\[paper\]](#) [\[code\]](#)
4. **J. M. Álvarez**, A. Mastropietro, and S. Ruggieri. The initial screening order problem. In *WSDM*, pages 165–174. ACM, 2025. [\[paper\]](#) [\[code\]](#)
5. **J. M. Álvarez** and S. Ruggieri. Counterfactual situation testing: Uncovering discrimination under fairness given the difference. In *EAAMO*, pages 2:1–2:11. ACM, 2023. [\[paper\]](#) [\[code\]](#)
6. **J. M. Álvarez***, K. M. Scott*, B. Berendt, and S. Ruggieri. Domain adaptive decision trees: Implications for accuracy and fairness. In *FAccT*, pages 423–433. ACM, 2023. [\[paper\]](#) [\[code\]](#)
7. C. Mougán, **J. M. Álvarez**, S. Ruggieri, and S. Staab. Fairness implications of encoding protected categorical attributes. In *AIES*, pages 454–465. ACM, 2023. [\[paper\]](#) [\[code\]](#)
8. S. Ruggieri, **J. M. Álvarez**, A. Pugnana, L. State, and F. Turini. Can we trust fair-AI? In *AAAI*, pages 15421–15430. AAAI Press, 2023. [\[paper\]](#)

Working Papers

1. Count Time Series Change Point Detection with W. Verbeke, J. Davis. *In preparation*.
2. The Weighted Partial Dependence Plot with A. Pugnana, S. Ruggieri. *In preparation*.

Workshops

1. **J. M. Álvarez**, A. Bringas-Colmenarejo, A. Elobaid, S. Fabbriizzi, M. Fahimi, A. Ferrara, S. Ghodsi, C. Mougán, I. Papageorgiou, P. R. Lobo, M. Russo, K. M. Scott, L. State, X. Zhao, and S. Ruggieri. Policy advice and best practices on bias and fairness in artificial intelligence. In *EWAF*, volume 3908 of *CEUR Workshop Proceedings*. CEUR-WS.org, 2024.

Editor

1. **J. M. Álvarez**, A. Fabris, C. Heitz, C. Hertweck, M. Loi, and M. Zehlike, editors. *Proceedings of the 2nd European Workshop on Algorithmic Fairness, Winterthur, Switzerland, June 7th to 9th, 2023*, volume 3442 of *CEUR Workshop Proceedings*, 2023. CEUR-WS.org

RESEARCH VISITS

KBC , Advanced Data Analytics and Modeling <i>Host</i> : Dr. Barak Chizi	Nov 2024 - Feb 2026 Leuven, BE
SCHUFA Holding AG , Innovation & Research Team <i>Host</i> : Dr. Klaus Broelemann	Jul - Aug 2023 Wiesbaden, DE
Saarland University , Probabilistic Machine Learning Group <i>Host</i> : Prof. Isabel Valera	Aug 2023 Saarbrücken, DE
KU Leuven , Declarative Languages and Artificial Intelligence Group <i>Host</i> : Prof. Bettina Berendt	Sep - Nov 2022 Leuven, BE

EXPERIENCE

KU Leuven, Computer Science Department, DTAI Lab Nov 2024 - Feb 2026
Postdoctoral Researcher (Mentors: [Jesse Davis](#), [Wouter Verbeke](#)) Leuven, BE

- Led research project at the intersection of user intent modeling over time and time series change point detection methods, with applications to the KBC Group

University of Pisa, Computer Science Department, KDD Lab Nov 2020 - Oct 2024
Research Fellow Pisa, IT

- Worked on algorithmic discrimination detection methods based on fair counterfactual reasoning, including a set of deliverables for the European Commission

Deloitte May 2017 - Oct 2020
Senior Consultant in Risk Modeling Brussels, BE

- Designed, developed, and deployed statistical machine learning models on Python and R for Benelux clients in the energy (e.g., predictive maintenance) and financial (e.g., loss given default) sectors

UK Government, Cabinet Office Jun - Aug 2016
Summer Economist London, UK

- Provided insights via statistical analysis to policy makers on a range of topics, such as studying the volunteering determinants in the UK and the variation in electoral costs across local authorities

Frontera Energy Jun - Aug 2013
Summer Strategic & Commercial Analyst Bogota, CO

- Elaborated daily oil and gas price reports, providing economic insight into the effects of current events on price fluctuations in the world markets for briefings and presentations

PARTICIPATION

Invited talk, Seminario del Centro de Analítica para Políticas Públicas (CAPP)	Nov 2025
Poster, AAAI/ACM AIES	Oct 2025
Invited talk, DTAI Seminar, KU Leuven	Oct 2025
Poster, SMiLe	Jun 2025
Invited talk, Discrimination, Fairness, and Algorithms Series, University of Mainz	Apr 2025
Poster, ACM WSDM	Mar 2025
Invited talk, Causality Reading Club, University of Pisa	Jul 2024
Contributed talk, 3rd European Workshop on Algorithmic Fairness	Jul 2024
Invited talk, EAAMO Bridges, ALP Working Group	May 2024
Invited talk, ASR Group, University of Oslo	Nov 2023
Poster, ACM EAAMO	Oct 2023
Doctoral colloquium, ACM EAAMO	Oct 2023
Contributed talk, ECML Workshop on Bias and Fairness in AI	Sep 2023
Contributed talk, ACM FAccT	Jun 2023
Poster, IMS International Conference on Statistics and Data Science	Dec 2022
Poster, NeurIPS Workshop on Algorithmic Fairness through the Lens of Causality	Dec 2022
Invited talk, 1st Comète Workshop on Ethical AI, Inria Polytechnique	Sep 2022
Doctoral colloquium, ACM FAccT	Jun 2022
Contributed talk, 1st European Workshop on Algorithmic Fairness	Jun 2022

SERVICE

- 2026** ECML (Reviewer)
4th ECML Workshop on Hybrid Human-Machine Learning and Decision Making (Organizer)
- 2025** AISTAT, ICML, ECML BIAS Workshop (Reviewer)
4th European Workshop on Algorithmic Fairness (Area Chair)
- 2024** NeurIPS, ACM FAccT, ICLR, ICML, ECML, ACM AIES (Reviewer)
3rd European Workshop on Algorithmic Fairness (Area Chair, Social Media Chair)
- 2023** NeurIPS, ACM FAccT, NeurIPS AFT Workshop (Reviewer)
2nd European Workshop on Algorithmic Fairness (Organizer, Social Media Chair)
- 2022** TheWebConf, NeurIPS AFCEP Workshop (Reviewer)
2nd ACM EAAMO (Social Media Chair)

European Association for Algorithmic Fairness, *Founding Board Member* [\[Link\]](#) 2023 - 2025
Supervised the yearly European Workshop on Algorithmic Fairness

The TSEconomist: Student Magazine, *Deputy Editor* [\[Link\]](#) 2016 - 2017
Led a team of 20 copy editors and writers, producing three issues

SKILLS

- Programming** Python, R, PySpark; past experience in Scala
- Tools** Git, L^AT_EX, Bash; past experience in Docker, Stata, QGIS
- Languages** English (fluent), Spanish (native), Italian (working proficiency),
French (limited proficiency)

Last updated: July 29, 2026